

AICHU TAN

Data Scientist | Machine Learning | AI Engineer

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PROFESSIONAL SUMMARY

Data Scientist with hands-on experience developing machine learning, statistical modeling, and AI applications across public health, agriculture, and urban analytics. Experienced building end-to-end data pipelines, predictive models, computer vision systems, and LLM-powered applications using Python, SQL, TensorFlow, LangChain, AWS, and Streamlit. Passionate about transforming complex data into actionable business insights through scalable, production-ready solutions.

TECHNICAL SKILLS

Programming: Python, SQL, R, JavaScript, C++

Data Analysis & Visualization: Pandas, NumPy, Data Cleaning, Data Wrangling, Exploratory Data Analysis (EDA), Statistical Analysis, Feature Engineering, Tableau, Matplotlib, Streamlit, ArcGIS

SQL & Databases: Advanced SQL (Joins, CTEs, Window Functions), ETL, Data Modeling

Machine Learning: Scikit-learn, XGBoost, Random Forest, Classification, Regression, Clustering, K-Prototypes, Cross-Validation, Hyperparameter Tuning

Deep Learning & AI: TensorFlow, LSTM, YOLOv8, Computer Vision, LangChain, Retrieval-Augmented Generation (RAG)

Time Series Forecasting: ARIMA, SARIMA, SARIMAX, Exponential Smoothing, Forecasting

Cloud & MLOps: AWS (EC2, S3, RDS), MLflow, Git, Hugging Face Spaces, Supabase, Render

PROFESSIONAL EXPERIENCE

Data Scientist

Sep 2025 - Present

SDSU DiMoLab

- Designed end-to-end SQL and Python ETL pipelines to ingest and standardize 27+ years of longitudinal meteorological, entomological, and population data for outbreak modeling.
- Developed automated data-quality validation rules to flag anomalies and filter reporting noise before downstream modeling.
- Diagnosed instability in daily-frequency case counts and re-aggregated data into 3-day and 5-day windows, stabilizing forecasts and achieving **R² up to 0.89**, enabling earlier, more targeted public health response.
- Co-authoring a research manuscript on outbreak forecasting for submission to a peer-reviewed journal.

AI Engineer

Feb 2025 - Dec 2025

SDSU Metabolism of Cities Living Lab

- Designed and shipped an AI-powered plant disease detection and advisory system (YOLOv8 + LangChain RAG), deployed publicly via Streamlit and Hugging Face Spaces.
- Diagnosed misclassification patterns across 35,000 plant disease images using confusion matrix analysis, then overhauled annotation quality control on 13,500+ images, scaling model performance from **70% to 89.4% mAP@50**.
- Built an LLM-powered retrieval system grounded in agricultural research literature to minimize hallucination and deliver actionable, farmer-facing recommendations; redesigned system architecture to reduce API cost.
- Co-authored research accepted at the 4th International One Health Conference (Rome, 2025).

SELECTED DATA SCIENCE PROJECTS

Predicting Restaurant Ratings Using Yelp Metadata

Python • SQL • XGBoost • Random Forest • MLflow

- Developed an end-to-end machine learning classification pipeline using 36,261 Yelp restaurant records to predict high- versus low-rated businesses based on operational attributes rather than review text.
- Engineered 61 predictive features from business metadata, including operating hours, service offerings, and accessibility; achieved **73.3% accuracy** and **0.79 ROC-AUC** using XGBoost, with experiment tracking in MLflow.
- Found via K-Prototypes clustering to identify customer behavior patterns, revealing that accessibility features were stronger predictors of restaurant ratings than price tier, providing actionable business insights.

Online Ticket Reservation System

MySQL • Python • AWS RDS

- Designed a normalized relational database (ERD → schema) with trigger-based validation to guarantee transactional data integrity across bookings and payments.
- Deployed the database on AWS RDS with role-based access control, enabling secure and scalable data management.
- Built a Python/Datapane dashboard surfacing **\$70K+** in revenue trends and customer behavior patterns for business decision-making.

Mapping for Change - Urban Inequality Analysis

Spatial Analysis • Time Series • Data Visualization

- Integrated five public datasets (ACS, 311, PITC, and Tree Equity Score) into an interactive ArcGIS dashboard to analyze housing burden, homelessness, and environmental equity across urban neighborhoods.
 - Performed exploratory data analysis and spatial statistical analysis to identify significant geographic clusters linking housing cost burden with reduced tree equity, supporting targeted policy recommendations.
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OTHER EXPERIENCE

Business Owner - Wholesale Clothing

Apr 2009 - Oct 2016

- Increased annual sales **50%** through customer segmentation, market analysis, and demand forecasting.
- Optimized inventory planning and supply chain operations using sales forecasting and purchasing analytics.
- Managed international sourcing and import/export operations across Taiwan, China, and Indonesia.

Process Engineer - ProMOS Technology

Jul 2004 - Feb 2009

- Applied statistical process control (SPC), root cause analysis, and data analysis to improve semiconductor manufacturing yield.
 - Reduced manufacturing defects through statistical modeling, process optimization, and cross-functional problem solving.
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RESEARCH & PUBLICATIONS

Climate-Driven Lag Modeling for Dengue Control Policy in Taiwan

Co-author • *Manuscript in preparation (2026)*

AI for Farmers: A YOLO and Language Model-Based Plant Disease Detection and Advisory System

Aichu Tan, Diego Vito, Gabriel Fernandez

Submitted to One Health Advances (Springer Nature); under peer review (2026)

Racemization and Hydrolysis of (S)-Naproxen 2,2,2-Trifluoroethyl Ester in Non-Polar Solvents

Journal of Physical Organic Chemistry, 2004 (Published under the name Dewi Hamza)

EDUCATION

M.S. in Big Data Analytics, San Diego State University - **GPA 4.0**

Aug 2024 - Dec 2025

Google Data Analytics Professional Certificate

Dec 2022

B.S. in Chemical Engineering, National Cheng Kung University (Taiwan)

Sept 1999 - June 2003